

DCASE 2021 CHALLENGE TASK1A TECHNICAL REPORT

Technical Report

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ABSTRACT

This report describes our method for Task 1a (Low-Complexity Acoustic Scene Classification with Multiple Devices) of the DCASE 2021 challenge. The task targets low complexity solutions for the classification problem. This report uses Residual Network (ResNet) model and uses Log Mel Spectrogram to process features. To compress system complexity, this report uses Post Training Static Quantization. Post Training Static Quantization are used to do the 8-bits quantization, this method can reduce the model size by four times. The accuracy of the method proposed in this report on the development data set is 73%, which is 25% higher than the baseline.

Index Terms— Low-Complexity Acoustic Scene Classification, Multiple devices, ResNet, Post Training Static Quantization.

1. INTRODUCTION

In our lives, sound carries a lot of information. We can judge the surrounding scene (acoustic scene) and what is happening (acoustic event) by the information carried by the sound we hear. In recent years, Deep neural networks have achieved the advanced results in many machines learning tasks [1]. This includes acoustic scene classification and acoustic event recognition. The accuracy of the judgment by the computer is gradually improving, even more than the human ear recognized. The development of algorithms that use computers to automatically extract this information has great potential in a variety of applications. For example, searching multimedia based on audio content, making context-aware mobile devices, intelligent monitoring systems that can sense hearing, robot hearing, unmanned cars. However, a lot of research is still needed to reliably identify sound scenes and sound events in real sound scenes. In a real acoustic environment, there are usually multiple sounds at the same time, and they will be disturbed by environmental noise. Queen Mary University of London (QMUL) organized the first DCASE (Challenge on Detection and Classification of Acoustic Scenes and Events) challenge in 2013 [2], paving the way for sound detection classifier performance evaluation. The DCASE

Challenge aims to expand the most advanced technical sound scene and event analysis methods,

As a routine task of the DCASE Challenge, Acoustic Scene Classification (ASC) is a task to classifying the recorded data to the place that the data was recorded. In this year's DCASE2021 challenge, Task1a targets low complexity solutions for the classification problem in term of model size, and uses audio recorded with multiple devices. Each data corresponds to one class out of ten, there is no data with multiple labels. The data is ten seconds, and the useful information is rarely.

The main problem of subtask A was to design a classifier that would work stably on a variety of microphone. However, the development data set mostly includes data collected from specific microphones, while the evaluation data will include data recorded using microphones that did not appear in the development set. And the model size should under 128 kilobytes. This translates into 32768 parameters when using float32 (32-bit float) which is often the default data type ($32768 \text{ parameter values} * 32 \text{ bits per parameter} / 8 \text{ bits per byte} = 131072 \text{ bytes} = 128 \text{ KB (kibibyte)}$). It is smaller than last year's limited.

This report builds a ResNet model for acoustic scene classification system. This report uses 11 layers ResNet model because the limited model size of the task. Because of the model size is very small, this report tests the appropriate parameters of frames and Mel bins by running comparative experiments. This report set five experiments, they are 28 frames 128 Mel bins, 28 frames 256 Mel bins, 43 frames 128 Mel bins, 43 frames 256 Mel bins and 28 frames 64 Mel bins. According to the results, the best parameters is 28 frames and 128 Mel bins. This report builds a small-size ResNet model at first, and then use post training static quantization method to compress the model. A model can be compressed to 1/4 of the original size by this method.

2. DATA PREPROCESSING

This section describes the methods about signal processing that this report to transform the audio sample into acoustic features.